

Self-Driving Car · Premium Course

3 live sessions · 2 hrs each · 11-15 yrs · from KES 9,000

Premium course outline **PREMIUM**

Overview

Build a chassis-based car that navigates on its own, finishing with a race day.

DETAIL

Track	Arduino Robotics & Electronics
Format	3 live sessions · 2 hrs each
Delivery	Live online (1:1 or small group) or in person, you choose when booking
Ages	11-15 yrs
Level	Intermediate → Advanced
Price	from KES 9,000
Maker Kit	Optional, KES 8,500 one-time: Arduino Uno R3 board + USB cable, Breadboard + 40 jumper wires, LEDs (red, yellow, green, white, RGB) + resistors, Push buttons, potentiometer and piezo buzzer, Light sensor (LDR) and sound sensor, HC-SR04 ultrasonic sensor + bracket, IR sensor, DHT11 temperature & humidity sensor, PIR motion sensor, SG90 servo motor, Relay module, 16x2 LCD screen (I2C), Battery holder, Storage box. Bought once and reused in every Arduino course. Needed at home for live online sessions; for in-person sessions we provide kits, so buy one only to keep building at home.
Parts pack	Optional, KES 2,500: 2WD car chassis with 2 DC motors and wheels, L298N motor driver, Second IR sensor, AA battery pack
Standalone	No
Prerequisites	Servo motor control (free)

Programme

- 1. Motorised Chassis.** Assemble the chassis and wire DC motors to an L298N motor driver, then program forward, reverse and turn controls and drive a test course.

2. **Line Follower.** Calibrate IR sensors on black vs. white, then program the car to follow a track on its own and tune its speed on the curves.
3. **Obstacle Avoidance.** Mount an HC-SR04 ultrasonic sensor so the car detects and steers around objects, then race the finished cars on race day!

What they take home

 Working self-driving car kit

Why parents love it

“The technology behind Waymo and Tesla — built from scratch by your 12-year-old.”

How to book

Places are limited and run on scheduled dates. To book or ask a question, email **tinkerwithanne@gmail.com** with the course name (**Self-Driving Car**), your child's age, and your preferred dates. Prices start from the amount above and may vary with group size and materials.